DataLens Pro: A Streamlit-Based Open-Source Business Intelligence Platform for Automated Dashboard Generation

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Abstract

Business Intelligence tools like Tableau and Power BI dominate enterprise analytics but impose significant licensing costs and vendor lock-in. This paper presents DataLens Pro, an open-source Streamlit application that automatically generates 50+ dashboards and 50+ visualizations from CSV/Excel uploads. The system employs automated column-type detection, dynamic filter propagation, and Plotly-based interactive charting to replicate core Tableau functionality. We implement a novel auto-dashboard pipeline that categorizes columns into dimensions and measures, then applies heuristic rules to instantiate KPI cards, distribution plots, correlation matrices, time-series analysis, and clustering views. A recently added HTML export feature enables sharing of static reports without Python dependencies. Benchmarks on a 1000-row sales dataset show sub-2-second rendering for all dashboards. The project is available at https://github.com/sayantanr/datalens-pro. DataLens Pro demonstrates that Python-based frameworks can deliver enterprise-grade BI capabilities while remaining fully extensible and cost-free.

Keywords: Business Intelligence, Data Visualization, Streamlit, Open Source, Automated Dashboards, Tableau Alternative

1 Introduction

The democratization of data analytics requires tools that balance power with accessibility. Commercial Business Intelligence platforms provide drag-and-drop dashboard creation but cost $70+ per user/month, creating barriers for students, researchers, and small organizations [1]. Meanwhile, Python data stacks offer flexibility but demand coding expertise for each visualization.

DataLens Pro bridges this gap by combining Streamlit's reactive web framework with automated visualization heuristics. Users upload structured data and receive an instant multi-page analytics suite without writing code. Unlike notebook-based tools, DataLens Pro provides a persistent UI with global filters, export capabilities, and a Tableau-like separation of dimensions and measures.

This paper makes three contributions: (1) An architecture for zero-config BI generation from tabular data, (2) Implementation of 50+ auto-dashboards covering descriptive, diagnostic, and basic predictive analytics, (3) A new HTML export module for stakeholder sharing. We evaluate the system on synthetic e-commerce data and discuss limitations and future work.

2 Related Work

2.1 Commercial BI Platforms

Tableau pioneered visual analytics with its VizQL engine [3]. Power BI integrates tightly with Microsoft ecosystems. Both use columnar engines and optimized rendering but are closed-source. Licensing models penalize wide deployment.

2.2 Open-Source Alternatives

Apache Superset and Metabase provide SQL-first dashboards but require database setup and manual chart building. Evidence.dev uses Markdown + SQL but targets developers. Streamlit has emerged for custom data apps [2], yet no prior work auto-generates Tableau-style dashboards from raw files.

2.3 Automated Visualization

Voyager and DataVoyager [4] recommend charts based on data types. LUX [5] adds intent-driven suggestions in Jupyter. DataLens Pro extends this by packaging recommendations into full dashboards with cross-filtering, similar to Tableau Stories.

3 System Architecture

DataLens Pro follows a 4-stage pipeline: Ingest, Profile, Generate, Render.

3.1 Data Ingestion Layer

The system accepts CSV, XLSX, and XLS via Streamlit's `file_uploader`. Multiple files are concatenated with `pd.concat`, enabling partitioned data analysis. `@st.cache_data` prevents re-parsing on widget interaction.

3.2 Schema Profiling

Column typing drives all downstream logic. Algorithm 1 classifies columns:

```
def detect_column_types(df):
    dimensions, measures, dates = [], [], []
    for col in df.columns:
        if pd.api.types.is_numeric_dtype(df[col]):
            if df[col].nunique() < 20 and df[col].nunique() / len(df) < 0.05:
                dimensions.append(col) # Low-cardinality numeric
            else:
                measures.append(col)
        elif pd.api.types.is_datetime64_any_dtype(df[col]):
            dates.append(col)
```

```
        else:
            try:
                pd.to_datetime(df[col], errors='raise')
                df[col] = pd.to_datetime(df[col])
                dates.append(col)
            except:
                dimensions.append(col)
    return dimensions, measures, dates
```

Listing 1: Column Type Detection

This mirrors Tableau's blue/green pill concept. Date parsing attempts ISO and common formats.

3.3 Dashboard Generation Engine

For each profile, the engine instantiates templates:

1. **KPI Dashboard:** Sum and average for all measures using `st.metric`
2. **Distribution Suite:** Histograms + box plots per measure
3. **Correlation Module:** Pearson heatmap + scatter matrix if $|measures| \geq 2$
4. **Time Series:** Line/area charts for each measure vs first date column
5. **Categorical Breakdown:** Bar and pie charts for dimensions with < 50 unique values
6. **Advanced Analytics:** PCA and K-Means clustering on standardized measures

Total: 50+ dashboards defined procedurally. Each respects global filters applied via sidebar.

3.4 Rendering and Interaction

Plotly Express generates all charts for interactivity: hover, zoom, selection. Streamlit tabs segment the UI to avoid DOM overload. Global filters use `st.multiselect` and `st.slider`, with `apply_global_filters` performing boolean indexing.

3.5 HTML Export Module

A new feature addresses sharing limitations. Upon button click, the app captures current dashboard state and renders it to standalone HTML using Plotly's `to_html` with `include_plotlyjs='cdn'`.

```
def export_dashboard_html(fig_list, filename="dashboard.html"):
    html_parts = ["<html><head><title>DataLens Pro Export</title></head
    ><body>"]
    html_parts.append("<h1>DataLens Pro - Automated Report</h1>")
    for fig in fig_list:
        html_parts.append(fig.to_html(full_html=False, include_plotlyjs
    ='cdn'))
    html_parts.append("</body></html>")
    with open(filename, 'w') as f:
        f.write('\n'.join(html_parts))
```

```
    return filename
```

Listing 2: HTML Export Logic

The exported file contains all charts with full interactivity and requires no Python runtime, enabling email distribution or static hosting.

4 Implementation Details

4.1 Technology Stack

Component	Technology
Frontend Framework	Streamlit 1.35+
Visualization	Plotly 5.22+
Data Processing	Pandas 2.2+, NumPy 1.26+
Statistics	SciPy, Scikit-learn
File Parsing	OpenPyXL, xlrd
Deployment	Python 3.9+

Table 1: Technology Stack

4.2 Performance Optimization

Rendering 50+ Plotly figures risks browser lag. We mitigate via: (1) Lazy tab loading, (2) Sampling for scatter matrices when $n > 5000$, (3) Agg-first for time series: monthly binning before plotting. On a 1000×45 dataset, initial load completes in 1.8s on Intel i5, 16GB RAM.

4.3 Code Organization

The 550+ line `app.py` modularizes generators: `generate_kpi_dashboard`, `generate_correlation_dash` etc. This enables easy extension. Adding a new dashboard requires 1 function + tab registration.

5 Evaluation

5.1 Dataset

We generated a 1000-row synthetic e-commerce dataset with 45 attributes using Faker and NumPy. Features include Order_Date, Region, Category, Gross_Sales, Profit, Customer_Satisfaction. Realistic correlations were injected: VIP segment has higher Unit_Price, weekends show different Channel distribution.

5.2 Functional Comparison with Tableau

Feature	Tableau Public	DataLens Pro
Cost	Free with limits	Free, MIT License
Auto Dashboards	Manual	50+ automatic
Data Limit	15M rows	Memory-bound
Coding Required	No	No
Extensibility	Limited	Full Python
HTML Export	Yes	Yes (new)
Row-level Security	Yes	No

Table 2: Feature Comparison

DataLens Pro achieves 80% of Tableau's self-service use cases without licensing. Limitations include absence of LOD expressions and published data sources.

5.3 User Study

5 CSE undergraduates from GCECT tested the tool. All created dashboards from their project CSVs in ¡5 minutes. The HTML export was cited as critical for submitting reports to faculty without requiring Streamlit installation.

6 Conclusion and Future Work

DataLens Pro demonstrates that open-source Python can replicate core BI workflows. The HTML export feature closes the sharing gap with commercial tools. The codebase at https://github.com/sayantanr/datalens-pro has been starred by peers and used in college coursework.

Future work: (1) Add calculated fields via `df.eval`, (2) Implement drag-drop shelf UI using `streamlit-sortables`, (3) Support live SQL connections, (4) Add PDF export with `plotly.io`. We also plan to integrate LLMs for natural-language chart queries.

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